ChatGPT

Computational Methods for Deconvolution of Spatial ATAC-seq Data

Executive summary

The most important finding from the literature is that, as of **April 26, 2026**, I did **not identify a purpose-built deconvolution algorithm developed specifically for spot-based spatial ATAC-seq** in the primary and official sources reviewed. The clearest evidence is a 2025 Bioinformatics benchmark, which explicitly states that "no dedicated methods exist" for spatial chromatin accessibility deconvolution and then evaluates five top spatial transcriptomics deconvolution methods on simulated and targeted spatial chromatin accessibility data instead. In that benchmark, **Cell2location** and **RCTD** were the most robust methods on spatial ATAC-like data; **Tangram** was highly reference-sensitive; **SpatialDWLS** and **DestVI** trailed and were often close to baseline. The same study also found that **highly variable peaks outperform highly accessible peaks** for feature selection in spatial ATAC deconvolution. [1]

That means the field is currently in a **transfer-learning / adaptation phase** rather than a mature, modality-native phase. In practice today, if the goal is **spot-level cell-type proportion estimation from spatial ATAC**, the strongest evidence-supported starting points are **Cell2location** and **RCTD**, adapted to peak-count matrices and paired with a careful **scATAC-seq or multiome reference**. If the goal is broader method development, the most informative neighboring literatures are: **spatial transcriptomics deconvolution** for count models, spatial priors, and graph learning; **bulk ATAC deconvolution** for marker-peak engineering and chromatin-specific reference construction; and **adjacent spatial epigenomics methods** such as **SPEED** and **Descart** for sparsity reduction, denoising, and peak-level spatial heterogeneity analysis. [2]

A practical implication follows from this landscape: **the bottleneck is no longer only inference**, but also **feature harmonization, reference construction, and benchmarking design**. Spatial ATAC has many more measured features than spatial RNA, much stronger sparsity, and fewer gold-standard datasets with known spot composition. The best current development strategy is therefore to combine: **chromatin-aware feature selection** from the ATAC literature, **probabilistic count-based deconvolution** from the spatial transcriptomics literature, **spatial priors or graphs** to stabilize neighboring spots, and **atlas-based denoising/imputation** from the spatial epigenomics literature. [3]

What the current literature actually covers

Spatial chromatin accessibility methods now include both **single-cell or near-single-cell spatial tagging approaches** and **spot-based protocols** that aggregate multiple cells per location. The 2025 spatial ATAC deconvolution benchmark distinguishes these explicitly: **Slide-tags** provides direct single-nucleus barcoding in intact tissue, whereas **spot-based protocols** generate aggregate measurements that require computational deconvolution if the goal is spot-level cell composition. The same paper frames deconvolution as specifically relevant to the **spot-based** use case. [4]

A second important scope distinction is between **true deconvolution** and **adjacent tasks**. In this report, “deconvolution” means inferring one or more of: **cell-type proportions**, **cell-type-specific accessibility**, or **spot-level imputed profiles attributable to constituent cell types**. By that definition, tools such as **SPEED** and **Descart** are not deconvolution methods, but they are directly relevant because they address two of the largest blockers for spatial ATAC deconvolution: **extreme sparsity** and **informative feature discovery**. Likewise, **Tangram** is not only a deconvolution method in the strict mixture-model sense; it is also a mapping framework that can transfer **chromatin accessibility and motif information** from multimodal single-cell data into space, which is highly informative for future spatial ATAC designs. [5]

The field can therefore be thought of as three overlapping tiers: direct spot deconvolution for spatial ATAC; neighboring transcriptomics methods that are transferable; and chromatin-specific deconvolution or spatial epigenomics methods that solve pieces of the overall problem. That structure is reflected in the method map below, which synthesizes the reviewed sources. [6]

```
flowchart TD
    A[Spatial ATAC deconvolution problem] --> B[Directly applied today]
    A --> C[Spatial transcriptomics methods informing design]
    A --> D[Bulk ATAC and spatial epigenomics methods informing design]

    B --> B1[Cell2location]
    B --> B2[RCTD]
    B --> B3[Tangram]
    B --> B4[SpatialDWLS]
    B --> B5[DestVI]
    B --> B6[deconvATAC benchmark and simulation framework]

    C --> C1[Count-based probabilistic models]
    C --> C2[Graph and spatial-prior models]
    C --> C3[Multimodal integration models]

    D --> D1[Marker-peak and reference-profile methods]
    D --> D2[Deep learning on bulk ATAC]
    D --> D3[Denoising and spatially variable peak detection]
```

Methods directly applied to spatial ATAC-seq

The core direct literature is currently dominated by **adaptation rather than invention**. The 2025 **deconvATAC** study benchmarked **Cell2location, RCTD, Tangram, SpatialDWLS, and DestVI** on simulated spot-based chromatin accessibility data derived from **human heart** and **human cerebral cortex** 10x Multiome datasets, plus a **human metastatic melanoma Slide-tags** dataset aggregated into pseudo-spots. Evaluation used **RMSE** and **Jensen–Shannon divergence** for proportion accuracy, and also **NMI** and **F1** for dominant-type and rare-cell recovery. [7]

```
timeline
    title Methods most relevant to spatial ATAC deconvolution
```

```
    2020 : DeconPeaker
    2021 : RCTD
         : Tangram
         : SpatialDWLS
    2022 : Cell2location
         : DestVI
         : CARD
    2024 : EPIC-ATAC
         : STdGCN
         : Descart
    2025 : deconvATAC benchmark
         : CLPLS
         : DECA
         : SPEED
```

Comparative table of direct methods

Method	Inputs, model, assumptions	Preprocessing and outputs	Benchmarks, performance, runtime	Software and reproducibility	Practical assessment for spatial ATAC
Cell2location — Kleshchevnikov, Shmatko, Dann *et al.* (2022); applied to spatial ATAC in Ouologuem, Martens, Schaar, Shulman, Gagneur, Theis (2025). [8]	Originally takes **spatial transcriptomics counts + annotated scRNA-seq reference**, using a **Bayesian count model** that explains spot counts by mixtures of reference signatures while modeling technical effects and borrowing strength across locations. In the spatial ATAC benchmark, it was applied to **spot-by-peak ATAC matrices** with reference labels from dissociated multiome/ scATAC-style data. Assumes the reference spans the observed cell types and that overdispersed count structure is still useful on peak counts. [9]	For spatial ATAC adaptation, the benchmark harmonized feature spaces and compared **highly variable** versus **highly accessible** peaks, with highly variable peaks performing better overall. Outputs are posterior **cell abundances / proportions per spot**; in the original RNA setting, the paper also reports spatial **cell and mRNA abundance** estimates. [10]	In the direct spatial ATAC benchmark, Cell2location was one of the **two best-performing methods**, robust over changes in density, heterogeneity, and zonation. On the Russell melanoma pseudo-spot benchmark, **ATAC-based deconvolution was slightly worse than RNA-based deconvolution**, with **NMI 0.445 versus 0.462** for Cell2location; rare cell F1 was also lower for ATAC than RNA. No formal runtime was reported in the benchmark text I reviewed. [11]	Official **Python** repository, **Apache-2.0**; documentation, **Google Colab**, and a **Docker image** are advertised in the official repo. [12]	**Best current evidence-supported default for spot-based spatial ATAC**, especially in heterogeneous tissues. Main limitations are heavier compute, RNA-centric count assumptions, and lack of native cell type-specific accessibility reconstruction. [13]

Method	Inputs, model, assumptions	Preprocessing and outputs	Benchmarks, performance, runtime	Software and reproducibility	Practical assessment for spatial ATAC
RCTD / spacexr — Cable, Murray, Zou *et al.* (2022). [14]	Originally takes **spatial counts + annotated scRNA-seq reference** and uses a **probabilistic maximum-likelihood model** assuming **Poisson-distributed counts with a log-normal prior**, with platform-effect correction. In spatial ATAC adaptation, the same framework was applied to **spot peak counts** and ATAC-style references. Assumes spot mixtures can be captured by a relatively small number of reference types; the benchmark used `doublet_mode="full"`. [15]	Requires construction of `SpatialRNA` and `Reference` objects in the software; for ATAC adaptation this means peak-count matrices and a harmonized feature space. Outputs include **spot weights / proportions**, plus spot class and dominant types in other modes. [16]	In the spatial ATAC benchmark, RCTD was the other **top-tier method**, with performance close to Cell2location but somewhat weaker in more heterogeneous, zonated tissue. The original repo documents runtime: example datasets run in **<5 minutes**, and a **Slide-seq cerebellum dataset with ~3,000 genes and 11,000 pixels** runs in **~20 minutes on a 4-core laptop**. [17]	Official **R** package, **GPL-3.0**; GitHub + Bioconductor, extensive **vignettes**, example outputs, and documentation. [18]	**Very strong current baseline for spatial ATAC**, especially if you want a method that is easier to operationalize than Cell2location and has better documented runtime. Main limitation is the RNA-origin mixture structure and no native chromatin-specific output. [13]

Method	Inputs, model, assumptions	Preprocessing and outputs	Benchmarks, performance, runtime	Software and reproducibility	Practical assessment for spatial ATAC
Tangram — Biancalani, Scalia, Buffoni, Avasthi, Lu *et al.* (2021). [19]	Takes **single-cell and spatial data from the same tissue / anatomical region** with a shared feature set, and learns a **cell-to-spot probability matrix** by optimizing similarity between mapped single-cell profiles and spatial measurements. It can run at cell or cluster level. Assumes close tissue match and is highly sensitive to training-feature choice and reference composition. Tangram is also notable because the original paper explicitly used **multimodal SHARE-seq** to transfer **chromatin accessibility and TF motif scores** into space. [20]	Requires intersected feature sets and training-feature selection; the repo suggests top marker genes or all shared features. Outputs include a **cell-by-spot probability matrix** and projected gene-expression maps; conceptually, the same mapping can transfer multimodal chromatin signals when available in the reference. [21]	In the spatial ATAC benchmark, Tangram showed **variable performance**. It could look strong on the Russell targeted dataset, but the benchmark authors cautioned that this was likely helped by the unusually **small, composition-matched reference**; performance deteriorated with larger or more complex references. The official repo states that a **single P100 GPU maps ~50k cells in a few minutes**. [22]	Official **Python** repository, **BSD-3-Clause**; documentation, tutorial, releases, and environment file. [23]	**High-upside but high-variance** method for spatial ATAC. Mos useful when the reference is tightly matched and multimodal; less trustworthy than Cell2location/ RCTD as a general-purpose spot deconvolver. [13]

Method	Inputs, model, assumptions	Preprocessing and outputs	Benchmarks, performance, runtime	Software and reproducibility	Practical assessment for spatial ATAC
SpatialDWLS — Dong and Yuan (2021). [24]	Uses **cell-type signatures / markers** plus spatial measurements, with an **enrichment-filtering step** followed by **dampened weighted least squares**. Assumes that only a small number of cell types are present at each location and that marker specificity is good. In the ATAC benchmark, it was adapted to **TF-IDF-normalized ATAC data with LSI**. [25]	Required preprocessing differs by modality: in the benchmark, RNA used **log normalization + PCA**, while ATAC used **TF-IDF + LSI**; clustering used Leiden. Outputs are **cell-type proportions per spot**. [4]	In its original RNA paper, SpatialDWLS reported better **RMSE** and speed than MuSiC, RCTD, SPOTlight, and stereoscope on coarse-grained **seqFISH+** and showed correct anatomical localization on **mouse brain Visium**; one example oligodendrocyte RMSE was **0.03**. In direct spatial ATAC benchmarking, however, it was **close to baseline** and degraded more under heterogeneity/ zonation. The paper states it was faster than the compared methods, but the exact runtime values were not recoverable from the accessible text I reviewed. [26]	Implemented in **Giotto** (**R**, **MIT**), with vignettes and associated code/data released via GitHub and Zenodo. [27]	A good **lightweight baseline** and still informative as a design pattern because it shows the value of **type filtering before regression**. For spatial ATAC today, I would no choose it over Cell2location or RCTD unless compute and simplicity dominate the decision. [13]

Method	Inputs, model, assumptions	Preprocessing and outputs	Benchmarks, performance, runtime	Software and reproducibility	Practical assessment for spatial ATAC
DestVI — Lopez, Li, Keren-Shaul *et al.* (2022). [28]	A **two-stage VAE** model: a single-cell latent variable model plus a spatial latent variable model. It was designed for **spatial transcriptomics + scRNA-seq** and explicitly models **continuous sub-cell-type states**, not only discrete proportions. Assumes negative-binomial-like counts and that within-type state variation is important and learnable from the reference manifold. [29]	In the direct spatial ATAC benchmark, the authors used **scvi-tools** with **300 epochs for scLVM** and **2,000 epochs for stLVM**. Original outputs are **cell-type proportions**, latent cell states, and **cell-type-specific imputed expression** inside each spot. In spatial ATAC adaptation, the benchmark focused on proportion recovery. [30]	The original paper reported that DestVI outperformed existing methods for estimating **cell-type-specific gene expression within spots** on simulations, and applied it to **murine lymph node** and **mouse tumor** data. In the spatial ATAC benchmark, however, DestVI was **inconsistent** and often **only slightly better than baseline**, especially on targeted-data simulations. I did not identify a formal runtime claim in the sources reviewed, although the model benefits from GPU-enabled scvi-tools infrastructure. [31]	Available through **scvi-tools** (**Python**, **BSD-3-Clause**), with tutorials, a Dockerfile in the main package, and a separate **reproducibility repo + Zenodo archive** for the paper. [32]	Scientifically attractive if you need **cell-state-aware** estimation but currently **not a top recommendation for spatial ATAC proportion deconvolution** based on the direct benchmark evidence. [33]

Method	Inputs, model, assumptions	Preprocessing and outputs	Benchmarks, performance, runtime	Software and reproducibility	Practical assessment for spatial ATAC
deconvATAC — Ouologuem, Martens, Schaar, Shulman, Gagneur, Theis (2025). Not a deconvolution algorithm, but the key **benchmark and simulation framework** for this problem. [34]	Accepts **dissociated multiome** or **targeted spatial multiome** data to simulate matched RNA and ATAC pseudo-spots. It is built specifically to test how transcriptomics deconvolution methods transfer to spatial chromatin accessibility. [34]	Supports feature selection comparisons, spot simulation under different densities/ zonation patterns, and evaluation metrics including **RMSE, JSD, NMI, and F1**. [35]	Benchmarked on **human heart multiome**, **human cerebral cortex multiome**, and **human metastatic melanoma Slide-tags** aggregated into pseudo-spots. This is the most important current reproducible baseline for future method development. [36]	Official **Python/ Jupyter** repo, **BSD-3-Clause**, with docs, notebooks, scripts, tests, and Zenodo-linked data. [37]	If you are **developing** a spatial ATAC deconvolution method, start here. It is presently the bes available reference benchmark and simulation scaffold. [37]

The strongest methodological takeaway from the direct evidence is that **count-aware probabilistic approaches transfer better than expected to peak-count data**, but they still do not fully solve the spatial ATAC problem. Cell2location and RCTD remain robust despite ATAC sparsity, yet ATAC-based deconvolution still lags matched RNA-based deconvolution, especially for **rare cell types**. The likely reasons, identified explicitly by the benchmark authors, are the intrinsic sparsity of ATAC and the fact that these models were built for RNA rather than chromatin accessibility. [11]

The second major takeaway is that **reference design matters almost as much as model choice**. Tangram's behavior in particular shows that a model can look excellent in a tightly matched, small reference setting and then degrade sharply when the reference becomes larger or less composition-matched. For anyone developing a new spatial ATAC method, this means benchmarks absolutely must vary **reference size, reference mismatch, rare-cell abundance, density, and zonation**; otherwise, conclusions about "best" methods are unstable. [22]

Related methods that inform spatial ATAC deconvolution design

The spatial ATAC problem is not isolated. The most promising design ingredients come from two neighboring literatures: **spatial transcriptomics deconvolution** and **bulk ATAC deconvolution**. I focus here on methods that add something particularly useful for spatial ATAC, rather than attempting an exhaustive census of the much larger spatial transcriptomics field. [38]

Selected spatial transcriptomics methods that are especially informative

Method	Why it matters for spatial ATAC	Key details	Software and reproducibility
CARD — Ma and colleagues (2022). [39]	CARD is especially relevant because it makes **spatial correlation in cell-type composition** a first-class modeling object. Spatial ATAC data are even noisier than spatial RNA, so explicit spatial priors should be even more valuable. [40]	Inputs are **spatial transcriptomics + scRNA-seq reference**; the model estimates cell-type composition while accounting for **spatial correlation across locations**, and can also produce a refined spatial map. Conceptually, this is one of the clearest templates for adding explicit spatial regularization to future spot-based ATAC deconvolution. [40]	Official **R** package and documentation are available; the official pages confirm GitHub availability, though I did not verify the license type in the sources gathered for this report. [40]
STdGCN — Spatial Transcriptomics deconvolution using Graph Convolutional Networks (2024). [41]	STdGCN matters because it shows how to infuse **spot adjacency graphs** into deconvolution. For spatial ATAC, graph smoothing over neighboring spots is likely to be particularly powerful given sparse peak matrices. [42]	The abstract states that STdGCN integrates **scRNA-seq profiles** with **spatial localization information** using a **GCN**, and benchmarked favorably against **17 state-of-the-art methods** across platforms. This graph-based design is directly portable to spatial ATAC if the features are peaks, motifs, or latent accessibility components rather than genes. [42]	Official GitHub repository includes a **tutorial notebook**. The repo metadata I reviewed did not clearly expose the license type. [43]

Method	Why it matters for spatial ATAC	Key details	Software and reproducibility
CLPLS — graph contrastive learning + partial least squares (2025). [44]	CLPLS is the most directly relevant recent spatial transcriptomics method for spatial ATAC because it was explicitly motivated by the inability of earlier methods to use **single-cell multi-omics, including chromatin accessibility**. [45]	It combines **graph contrastive learning** with **PLS regression** for spot deconvolution, and the authors emphasize that it can be extended to integrate **single-cell multi-omics** so that epigenomic landscapes become accessible in spatial analysis. That makes it a strong conceptual precursor for a future spatial ATAC-native model. [45]	Official GitHub repository exists. In the sources I reviewed, I did not recover the full software license or detailed reproducibility assets beyond the repo itself. [46]
Stereoscope — Andersson and colleagues (2020). [47]	Stereoscope is less directly used in spatial ATAC today, but it remains a foundational **probabilistic reference-based deconvolution** model and thus helpful as a conceptual baseline. [48]	The official repo describes it as the implementation of **probabilistic inference of cell-type topography** by integrating single-cell and spatial transcriptomics. For spatial ATAC, the primary lesson is still the value of explicit generative count models before adding more elaborate deep-learning layers. [47]	Official **Python** package, **MIT** license, with example scripts used for preprocessing, visualization, and comparison. [47]

Selected bulk ATAC deconvolution methods that are especially informative

Method	Core idea and inputs	Outputs, performance, limitations	Software and reproducibility
DeconPeaker — Li, Sharma, Luo, Qin, Sun, Liu (2020). [49]	Designed specifically for **bulk ATAC-seq mixtures**. It identifies **cell-type-specific peaks (CTSPs)**, constructs a signature matrix with a **condition-number minimization** strategy, and performs deconvolution with **SIMPLS**, a partial least squares method. Inputs are purified/reference ATAC profiles or synthetic references; preprocessing includes MACS2 peak calling, blacklist filtering, fragment counting, quantile normalization, and CTSP selection. [50]	Outputs are **cell-type proportions** and a consistency test. On the authors' benchmarks, DeconPeaker reported **lowest average RMSE = 0.042** and **highest average correlation r = 0.919** relative to known proportions in comparison with alternative methods; it also showed better PCC and lower RMSE than CIBERSORT on synthetic chromatin-accessibility mixtures. A repo demo reports an example elapsed time of **~29.6 seconds**. A limitation for spatial ATAC is that it is not spatially aware and was optimized for bulk-style mixtures. [51]	Official implementation is a **Python/R mix** with a **demo notebook** and tutorial site. I did **not** verify an explicit OSI license in the accessible repo metadata. [52]

Method	Core idea and inputs	Outputs, performance, limitations	Software and reproducibility
EPIC-ATAC — Gabriel, Racle, Falquet, Jandus, Gfeller (2024). [53]	Extends the **EPIC** framework from RNA to **bulk tumor ATAC-seq**, using **marker peaks and reference ATAC profiles** for immune, stromal, endothelial, and "other" cells. Inputs are **TPM-like normalized ATAC counts**, with utilities to convert raw counts. A key design feature is the ability to estimate an **uncharacterized / other** cell component, important for tumors. [54]	Outputs are **cell-type proportions**, including an "other" compartment. The paper and associated resources report evaluation on **PBMC bulk ATAC with flow-cytometry ground truth**, multiple **scATAC-derived pseudobulks**, and a **human breast cancer cohort**, and conclude that EPIC-ATAC accurately predicts non-malignant and malignant fractions. From the sources reviewed here, I did not recover a compact numerical summary table analogous to DeconPeaker's. [55]	Official **R** package repo with tests and vignettes; a separate manuscript-analysis repo is also listed. A license file is present in the repo, but the precise license type was not visible in the accessible metadata I reviewed. [56]
DECA — Luo, Zhu, Lin *et al.* (2025). [57]	A **vision transformer + decoder** framework for **bulk ATAC deconvolution** from **scATAC references**, trained on pseudo-bulk mixtures generated from single-cell data. It explicitly predicts both **cell-type proportions** and **reconstructed chromatin accessibility matrices**. Training uses Dirichlet-like pseudo-mixtures and MSE-based objectives. [58]	Reported metrics include **Lin's CCC, MAE, and Spearman correlation**. The paper reports **Spearman 0.98** on PBMC simulation samples and, on an immune atlas analysis, **correlation 0.97**, **mean MAE 0.02**, and **mean CCC 0.95**. Runtime is described qualitatively as **moderate**: longer than TAPE, Cellformer, and Bisque, but shorter than DWLS. DECA is highly relevant to spatial ATAC because it demonstrates a chromatin-native deep architecture that reconstructs accessibility, not only proportions. [59]	Official **Python/Jupyter** repository with **Training.ipynb**, **Prediction.ipynb**, and **environment.yaml**. I did **not** verify an explicit license in the visible repo metadata. [60]

Method	Core idea and inputs	Outputs, performance, limitations	Software and reproducibility
DECODER — anchor-based regression package, repo version dated 2019. [61]	Anchor-based regression for **bulk ATAC deconvolution using scATAC references**. Inputs can be bulk samples plus a built-in hematopoietic reference or user-provided scATAC-derived reference; the package also exposes a **BAM → peak calling → counting → deconvolution** pipeline. [62]	Outputs are estimated **cell-type proportions**. In the sources gathered for this report, I did not locate a formal peer-reviewed paper with benchmarking results, so the empirical evidence is weaker than for DeconPeaker, EPIC-ATAC, or DECA. [63]	Official **R** package repository with demo data and extdata examples. I did **not** verify an explicit license in the visible repo metadata. [62]

Adjacent spatial epigenomics methods that are not deconvolution, but are highly relevant

Method	Relevance to spatial ATAC deconvolution
SPEED (2025) denoises spatial epigenomic data using **deep matrix factorization**, atlas-level single-cell epigenomic data, and **spatial context**. It is directly relevant because it attacks the **sparsity/noise regime** that undermines spot deconvolution, and the paper reports improved denoising, dimensionality reduction, differential accessibility analysis, and spatial-domain identification across spatial epigenomic modalities. [64]	
Descart (2024) detects **spatially variable chromatin accessibility peaks** using a graph of inter-cellular correlations. It is not a deconvolver, but it is highly relevant for **feature selection**, **evaluation**, and identifying peaks that carry both spatial and cell-type heterogeneity information. [65]	

Taken together, these neighboring methods suggest a very plausible blueprint for a future spatial ATAC-native deconvolution model: start with **chromatin-specific markers or learned peak embeddings** from bulk ATAC methods, stabilize the data using **atlas-guided denoising** from spatial epigenomics, then perform deconvolution with a **probabilistic or graph-regularized model** that outputs both **proportions** and **cell-type-specific accessibility reconstructions**. [66]

Benchmarking strategy, method selection, and development recommendations

If the task today is **estimating spot-level cell-type proportions from spatial ATAC**, the evidence-based ranking is straightforward. **Cell2location** is the strongest default when you can tolerate heavier modeling and want the most robust performance in heterogeneous tissue. **RCTD** is the strongest simpler alternative and has better documented runtime characteristics. **Tangram** is worth trying when you have an unusually well-matched and possibly multimodal reference, but I would treat it as a high-variance option rather than a primary baseline. **SpatialDWLS** is a sensible lightweight baseline. **DestVI** should be reserved for cases where continuous cell-state structure is central and you have a strong validation plan, because its direct spatial ATAC performance is currently weak. [67]

For **preprocessing**, the current best-supported recommendation is to build a **shared peak space** between spatial ATAC and the reference, then select **highly variable peaks rather than merely highly accessible peaks**. The benchmark found the highly variable strategy consistently superior, with a reported **P = 0.007**. For methods that were built for RNA, peak-level feature harmonization is not a side issue; it is core modeling infrastructure. Newer work on **consensus peak references** for single-cell ATAC is therefore relevant even though it is not itself a deconvolution method, because it may reduce one of the most important sources of cross-dataset mismatch. [68]

For **benchmark datasets**, the minimum credible battery should include both **silver-standard pseudo-spots** and **gold-standard or near-gold-standard spatial datasets**. The best current starting set is the exact combination used in deconvATAC: **human heart** and **human cerebral cortex** multiome datasets for controlled pseudo-spot simulations, plus the **human metastatic melanoma Slide-tags** dataset aggregated into artificial spots for a more realistic spatial benchmark. For future work, I would add newer spatial chromatin datasets such as **spatial FFPE-ATAC** and **SPACE-seq** once sufficiently matched single-cell or orthogonal composition labels are available, because they broaden tissue preservation and chemistry conditions. [69]

For **evaluation metrics**, the current literature strongly supports a multi-axis view. At minimum, use **RMSE** and **JSD** for composition accuracy; **NMI** for dominant-type localization; and **F1** for rare cell recovery, because spatial ATAC is weakest exactly where rare types matter. To this I would add **MAE, Spearman/Pearson, and Lin's CCC** when pseudo-bulk ground truth is available, drawing on the bulk ATAC literature. For Bayesian methods, I recommend adding **uncertainty calibration and posterior coverage**, even though most published spatial ATAC evaluations have not yet made that a standard endpoint. [70]

The main **gaps and open challenges** are now clear. First, the field still lacks a **modality-native spatial ATAC deconvolution model** with an ATAC-appropriate likelihood and chromatin-aware outputs. Second, there is no widely accepted benchmark for **unknown / novel cell states** or for **missing-reference** conditions, even though these are common in disease tissues. Third, there is still very limited ground truth for **cell-type-specific accessibility profiles in space**, meaning most evaluations stop at proportions instead of the richer outputs that chromatin data should enable. Fourth, scaling remains an issue, because spatial ATAC has far more features than spatial RNA, and naïve transfer of RNA tools can become unstable or reference-sensitive. [71]

My recommended design for a **next-generation spatial ATAC deconvolution method** would combine four ingredients. First, an **ATAC-native shared feature space**, ideally based on consensus peak references and explicit marker-peak engineering. Second, a **spot-level generative mixture model** that allows zero inflation / sparsity-aware count behavior and includes an **unknown-cell or residual component** in the spirit of EPIC-ATAC. Third, a **spatial regularization layer** using graphs or Gaussian-process-style priors. Fourth, a **decoder for cell-type-specific accessibility reconstruction**, as already demonstrated in the bulk

ATAC setting by DECA. That combination would directly address the main weaknesses identified by current benchmarks. [72]

Open questions and limitations

A few parts of the evidence base remain incomplete. I did **not** find a post-2025, purpose-built spatial ATAC deconvolution method in the sources reviewed, but absence of evidence is not a mathematical proof of absence; it is a high-confidence literature finding from this source set. Likewise, for some neighboring methods, especially **CLPLS**, **CARD**, and **DECODER**, I verified the core algorithmic idea and software existence but did **not** recover a complete, paper-level set of runtime, license, or benchmark details from the accessible sources gathered here. I have marked those cases explicitly rather than filling gaps speculatively. [73]

There is also a structural limitation in the field itself: many of the most important direct comparisons for spatial ATAC are still mediated through **simulations and pseudo-spots**, because true gold-standard datasets with both spatial chromatin accessibility and known spot composition remain sparse. That does not invalidate the current conclusions, but it means any ranking of methods should still be treated as **provisional and benchmark-dependent**. [74]

Prioritized sources

The papers and repositories below are the highest-priority sources for this topic.

Direct spatial ATAC deconvolution evidence

- **Ouologuem S, Martens LD, Schaar AC, Shulman M, Gagneur J, Theis FJ.** *Spatial transcriptomics deconvolution methods generalize well to spatial chromatin accessibility data.* Bioinformatics, 2025. This is the central direct benchmark and simulation framework for the problem. [4]
- **theislab/deconvATAC** official repository. Contains the benchmark implementation, simulations, notebooks, scripts, and docs. [37]

Methods directly applied to spatial ATAC

- **Kleshchevnikov V, Shmatko A, Dann E, et al.** *Cell2location maps fine-grained cell types in spatial transcriptomics.* Nature Biotechnology, 2022. [75]
- **BayraktarLab/cell2location** official repository. [76]
- **Cable DM, Murray E, Zou LS, et al.** *Robust decomposition of cell type mixtures in spatial transcriptomics.* Nature Biotechnology, 2022. [77]
- **dmcable/spacexr** official repository for RCTD/C-SIDE. [78]
- **Biancalani T, Scalia G, Buffoni L, et al.** *Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram.* Nature Methods, 2021. [79]
- **broadinstitute/Tangram** official repository. [80]
- **Dong R, Yuan G-C.** *SpatialDWLS: accurate deconvolution of spatial transcriptomic data.* Genome Biology, 2021. [81]
- **giotto-suite/Giotto** official repository containing SpatialDWLS functionality. [82]
- **Lopez R, Li B, Keren-Shaul H, et al.** *DestVI identifies continuums of cell types in spatial transcriptomics data.* Nature Biotechnology, 2022. [83]

- **scverse/scvi-tools** official repository; **DestVI-reproducibility** is linked from the paper. [84]

Related methods informing spatial ATAC design

- **Ma and colleagues.** *CARD* and official repository / docs for spatially informed deconvolution. [40]
- **STdGCN** paper and official repository. [41]
- **CLPLS** paper and official repository. [85]
- **Li H, Sharma A, Luo K, Qin ZS, Sun X, Liu H.** *DeconPeaker.* Frontiers in Genetics, 2020. [86]
- **lihuamei/DeconPeaker** official repository. [87]
- **Gabriel AAG, Racle J, Falquet M, Jandus C, Gfeller D.** *Robust estimation of cancer and immune cell-type proportions from bulk tumor ATAC-Seq data.* eLife, 2024. [88]
- **GfellerLab/EPIC-ATAC** official repository. [89]
- **Luo S, Zhu M, Lin L, et al.** *DECA: harnessing interpretable transformer model for cellular deconvolution of chromatin accessibility profile.* Briefings in Bioinformatics, 2025. [90]
- **xmuhuanglab/DECA** official repository. [91]
- **SPEED** paper on deep matrix-factorization denoising for spatial epigenomics. [64]
- **Descart** paper on spatially variable chromatin accessibility peak detection. [65]

---

[1] [2] [3] [4] [6] [7] [10] [11] [13] [15] [17] [22] [30] [33] [34] [35] [36] [67] [68] [69] [70] [71] [74] https://academic.oup.com/bioinformatics/article/41/Supplement_1/i314/8199385
https://academic.oup.com/bioinformatics/article/41/Supplement_1/i314/8199385

[5] [64] [66] https://www.nature.com/articles/s43588-025-00941-3
https://www.nature.com/articles/s43588-025-00941-3

[8] [9] [75] https://www.nature.com/articles/s41587-021-01139-4
https://www.nature.com/articles/s41587-021-01139-4

[12] [76] https://github.com/BayraktarLab/cell2location
https://github.com/BayraktarLab/cell2location

[14] [77] https://www.nature.com/articles/s41587-021-00830-w
https://www.nature.com/articles/s41587-021-00830-w

[16] [18] [78] https://github.com/dmcable/spacexr
https://github.com/dmcable/spacexr

[19] [20] [21] [23] [80] https://github.com/broadinstitute/Tangram
https://github.com/broadinstitute/Tangram

[24] [25] [26] [81] https://link.springer.com/article/10.1186/s13059-021-02362-7
https://link.springer.com/article/10.1186/s13059-021-02362-7

[27] [82] https://github.com/giotto-suite/Giotto
https://github.com/giotto-suite/Giotto

[28] [29] [31] [83] https://www.nature.com/articles/s41587-022-01272-8
https://www.nature.com/articles/s41587-022-01272-8

[32] [84] https://github.com/scverse/scvi-tools
https://github.com/scverse/scvi-tools

[37] https://github.com/theislab/deconvATAC
https://github.com/theislab/deconvATAC

[38] https://www.nature.com/articles/s41576-025-00845-y
https://www.nature.com/articles/s41576-025-00845-y

[39] https://www.nature.com/articles/s41587-022-01273-7
https://www.nature.com/articles/s41587-022-01273-7

[40] https://github.com/YMa-lab/CARD
https://github.com/YMa-lab/CARD

[41] [42] https://link.springer.com/article/10.1186/s13059-024-03353-0
https://link.springer.com/article/10.1186/s13059-024-03353-0

[43] https://github.com/luoyuanlab/stdgcn
https://github.com/luoyuanlab/stdgcn

[44] [45] [73] [85] https://academic.oup.com/bib/article/26/1/bbaf052/8006047
https://academic.oup.com/bib/article/26/1/bbaf052/8006047

[46] https://github.com/lhzhanglabtools/CLPLS
https://github.com/lhzhanglabtools/CLPLS

[47] [48] https://github.com/almaan/stereoscope
https://github.com/almaan/stereoscope

[49] [50] [51] [86] https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2020.00392/full
https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2020.00392/full

[52] [87] https://github.com/lihuamei/DeconPeaker
https://github.com/lihuamei/DeconPeaker

[53] [88] https://elifesciences.org/articles/94833
https://elifesciences.org/articles/94833

[54] [55] [56] [89] https://github.com/GfellerLab/EPIC-ATAC
https://github.com/GfellerLab/EPIC-ATAC

[57] [58] [59] [90] https://academic.oup.com/bib/article/26/1/bbaf069/8030609
https://academic.oup.com/bib/article/26/1/bbaf069/8030609

[60] [91] https://github.com/xmuhuanglab/DECA
https://github.com/xmuhuanglab/DECA

[61] https://github.com/jwyang16/DECODER/blob/master/DESCRIPTION
https://github.com/jwyang16/DECODER/blob/master/DESCRIPTION

[62] [63] https://github.com/jwyang16/DECODER/
https://github.com/jwyang16/DECODER/

[65] https://link.springer.com/article/10.1186/s13059-024-03458-6
https://link.springer.com/article/10.1186/s13059-024-03458-6

[72] https://www.nature.com/articles/s41467-026-69461-6
https://www.nature.com/articles/s41467-026-69461-6

79 https://hscrb.harvard.edu/publication/deep-learning-and-alignment-of-spatially-resolved-single-cell-transcriptomes-with-tangram-nat-methods-2021-oct-28/
https://hscrb.harvard.edu/publication/deep-learning-and-alignment-of-spatially-resolved-single-cell-transcriptomes-with-tangram-nat-methods-2021-oct-28/